



# CAPABILITY STATEMENT

## COMPANY OVERVIEW

AI Security Assurance is on a mission to accelerate delivery of trustworthy systems. We focus on automated, AI-driven secure by design systems analysis and assurance, including software. Grounded in systems theory and formal methods, our tools accelerate the delivery of trustworthy systems where the cost of a hidden design flaw is catastrophic. We are a fully remote, EDWOSB-certified small business with key personnel located in the DC-Maryland-Virginia region and experience spanning cybersecurity, formal methods, and quantum algorithms.

## AI Security Assurance, LLC

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## CORE COMPETENCIES

- Automated systems security and safety analysis (STPA-Sec, STAMP-grounded)
- Software architecture recovery
- Formal verification of security-critical properties
- AI/ML engineering for assurance tooling
- Methodology generalization (STPA+)
- Quantum algorithms, applied computational mathematics, graph theory, combinatorics

## NAICS CODES

- 541715 – Research and Development in the Physical, Engineering, and Life Sciences
- 513210 – Software Publishers
- 541511 – Custom Computer Programming Services
- 541519 – Other Computer Related Services
- 541512 – Computer Systems Design Services

## PAST PERFORMANCE

We are a startup, but come with a strong team of experts with degrees and experience in computer science, cybersecurity, AI/ML, software engineering, formal methods, quantum algorithms, graph theory & combinatorics, and applied computational mathematics. Team members have experience in academia, industry, and government. We're building our flagship product and seeking government funding to transition from concept to a dual-use capability that accelerates delivery of trustworthy systems.

## CERTIFICATIONS & MEMBERSHIP

- EDWOSB (Economically Disadvantaged Woman-Owned Small Business) – certified with SBA
- Virginia Apex Accelerator
- Catalyst Center for Business & Entrepreneurship

## DIFFERENTIATORS

- Direct STAMP/STPA methodology lineage, founding our products on trusted methodologies.
- Generalization of STPA to any system or assurance goal.
- Addressing OWASP Insecure Design problem.
- Tooling built around the tools you use: MBSE, MBSD, SysML v2, STPA, formal methods
- EDWOSB triple set-aside
- Highly educated and experienced team

## FACILITIES & EQUIPMENT

- **Fully remote workforce** with key personnel in the DC-Maryland-Virginia region.
- **Development infrastructure:** Self-managed GPU resources (NVIDIA Spark / DGX-class developer hardware) and cloud-hosted inference via enrolled AWS, Google Cloud, and NVIDIA startup programs.
- **Cleared personnel:** Founder holds active Secret clearance.
- Working towards CMMC Level 1 certification.